Day 051 | Unit 04 | Teacher plan

Numeric and Algebraic Expressions

Factor Sums with the Distributive Property

Standards: 6.NS.B.4c

Learning target: I can factor a sum using the greatest common factor.

enVision Topic 3: Numeric and Algebraic Expressions | Spectrum companion Chapter 1

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Find the GCF of the two addends. Divide each addend by that factor and place the quotients inside parentheses. The inside numbers should share no common factor greater than 1.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 17

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01 / Learn

Find the GCF of the two addends. Divide each addend by that factor and place the quotients inside parentheses. The inside numbers should share no common factor greater than 1.

Worked example 1

Factor 8 + 12 using its GCF. Make the inside numbers relatively prime.

1. Find the GCF of the two addends.

2. Divide each addend by that factor and place the quotients inside parentheses.

3. GCF = 4

4. 8 = 4 × 2 and 12 = 4 × 3.

5. State the result in the form requested: 4(2 + 3).

Answer: 4(2 + 3)

Worked example 2

Select all expressions equal to 36 + 18.
A. 6(6 + 3)
B. 55
C. 6 × 10
D. 36 + 18

1. Find the GCF of the two addends.

2. Distribute 6 to both terms: 6 × 6 + 6 × 3 = 54.

3. The other choices are 6 or 1 too large.

Answer: A, D - 6(6 + 3); 36 + 18

Day 051 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Factor 6 + 9 using its GCF. Make the inside numbers relatively prime.

02 / Constructed response

Factor 10 + 15 using its GCF. Make the inside numbers relatively prime.

03 / Constructed response

Factor 12 + 16 using its GCF. Make the inside numbers relatively prime.

04 / Constructed response

Factor 28 + 42 using its GCF. Make the inside numbers relatively prime.

05 / Constructed response

Factor 48 + 72 using its GCF. Make the inside numbers relatively prime.

06 / Constructed response

Factor 12 + 18 using its GCF. Make the inside numbers relatively prime.

07 / Constructed response

Factor 18 + 30 using its GCF. Make the inside numbers relatively prime.

08 / Multiple select

Select all expressions equal to 6 + 6.
A. 13
B. 2(3 + 3)
C. 2 × 7
D. 6 + 6

09 / Multiple select

Select all expressions equal to 25 + 15.
A. 5 × 9
B. 5(5 + 3)
C. 41
D. 25 + 15

10 / Constructed response

Factor 70 + 100 using its GCF. Make the inside numbers relatively prime.

Day 051 | Unit 04 | Answer key

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1. 3(2 + 3)

GCF = 3; 6 = 3 × 2 and 9 = 3 × 3.

6.NS.B.4c | Constructed response

2. 5(2 + 3)

GCF = 5; 10 = 5 × 2 and 15 = 5 × 3.

6.NS.B.4c | Constructed response

3. 4(3 + 4)

GCF = 4; 12 = 4 × 3 and 16 = 4 × 4.

6.NS.B.4c | Constructed response

4. 14(2 + 3)

GCF = 14; 28 = 14 × 2 and 42 = 14 × 3.

6.NS.B.4c | Constructed response

5. 24(2 + 3)

GCF = 24; 48 = 24 × 2 and 72 = 24 × 3.

6.NS.B.4c | Constructed response

6. 6(2 + 3)

GCF = 6; 12 = 6 × 2 and 18 = 6 × 3.

6.NS.B.4c | Constructed response

7. 6(3 + 5)

GCF = 6; 18 = 6 × 3 and 30 = 6 × 5.

6.NS.B.4c | Constructed response

8. B, D - 2(3 + 3); 6 + 6

Distribute 2 to both terms: 2 × 3 + 2 × 3 = 12. The other choices are 2 or 1 too large.

6.NS.B.4c | Multiple select

9. B, D - 5(5 + 3); 25 + 15

Distribute 5 to both terms: 5 × 5 + 5 × 3 = 40. The other choices are 5 or 1 too large.

6.NS.B.4c | Multiple select

10. 10(7 + 10)

GCF = 10; 70 = 10 × 7 and 100 = 10 × 10.

6.NS.B.4c | Constructed response